

Zhonghua YAO

Department of Earth and Planetary Sciences
The University of Hong Kong,
Pok Fu Lam, Hong Kong

Academic Qualifications

Ph.D. in Space Physics, Peking University, Beijing, China, January 2014
B.Sc. in Physics, Capital Normal University, Beijing, China, July 2008

Academic Appointments

2026 – Present, Director of HKU's Laboratory for Space Research (LSR), The University of Hong Kong
2023 – Present, Associate Professor, Department of Earth and Planetary Sciences, The University of Hong Kong
2024 – Present, Director and Research Professor, Planetary Space Environment Laboratory, The University of Hong Kong Shenzhen Institute of Research and Innovation (HKU-SIRI)
2019 – 2023: Research Professor, Institute of Geology and Geophysics, Chinese Academy of Sciences
2016 – 2019: Independent Researcher at University of Liège, Marie-Curie COFUND fellowship
2014 – 2016: Research Associate at Mullard Space Science Laboratory, University College London

Honorary Academic Appointments

2022 – Present: Honorary Professor, University College London (UK)
2024 – 2026: President of Planetary Science Section, Asia Oceania Geosciences Society (AOGS)

Grant records in Recent Five Years

1. **Principal Investigator**, *NSFC Young Scientists Fund (Type A)*, CNY 4,000,000.00, 2026
2. **Project Coordinator**, *Low Cost Cislunar (earth to moon) Transportation and Lunar Flash Observation (ITS/001/24SC)*, HK\$ 33,672,000.00, Innovation and Technology Support Programme Special Call (Aerospace Technology), since 2026
3. **Principal Investigator**, *The Effect of Jovian Magnetic Field Curvature on Particle Pitch-Angle Scattering*, HK\$ 1,279,308.00, General Research Fund (GRF), Research Grants Council (RGC), 2026
4. **Principal Investigator**, *Principal Scientific Research on Jupiter's Space Environment and Exploration Principles (#4)*, CNY 2,260,000.00, National Key Research & Development Programme, Ministry of Science and Technology (MOST), 2025
5. **Principal Investigator**, *The global picture and mechanisms of the periodic perturbations in Jupiter's magnetosphere-ionosphere system*, HK\$ 910,742.00, General Research Fund (GRF), Research Grants Council (RGC), 2024
6. **Principal Investigator**, *Comparative investigations of magnetospheric energy release and particle acceleration at planets*, CNY 530,000.00, General Research Fund, National Natural Science Foundation of China (NSFC), 2023

Involvements in Space Missions and Awards

1. Science Core Team Member in Tianwen-4 mission, 2026
2. Science Team Member in CE-8 X-ray Instrument, 2024
3. Science Team Member in Tianwen-2 Mission, 2024
4. Co-Investigator, XMM-Newton Obs., Are Jupiter's Pulses Signatures of Global or Local Wave-Particle Interactions? 2023
5. Co-Investigator, Hubble Space Telescope Obs., Building connections: Juno and STIS contemporaneous observations of Jupiter's magnetosphere and auroras, 2022

6. Co-Investigator, Chandra Obs., Chandra EUV Capabilities: A New View of Jupiter, 2021
7. Co-Investigator, XMM-Newton Obs., Juno and XMM-Newton's final search for drivers of Jupiter's X-ray aurorae, 2021
8. Kamide Lecture Award, AOGS, 2021
8. Cassini Group Achievement Award, NASA, 2018
9. Arne Richter Award, European Geosciences Union, 2018
10. Winton Capital Award, Royal Astronomical Society, 2017

Publication Record: <https://www.webofscience.com/wos/author/record/H-8537-2017>

209 publications, including 60 as first/correspondence author.

Selected publications as first or correspondence author

1. **Yao, Z.**, Guo, R., Zeng, Z., Zhang, B., Zhao, J., Sun, T., & Li, W. (2026). Quasi-periodic signatures of Jupiter's rotation in the tailside magnetosheath: Observations from Juno and Voyager-2 at ~ 100–200 Jupiter radii. *Science China Earth Sciences*, 1-6.
2. Sun, J. W., Xie, L.*, **Yao, Z. H.***, Fu, S. Y., Zhang, B., Guo, R. L., ... & Zeng, Z. L. (2026). A Heliosphere-like Current Sheet in Jupiter's Magnetosphere with Hinging Geometry. *The Astrophysical Journal*, 1007(1), 70.
3. Tian, S.*, **Yao, Z.***, Wygant, J.R. et al. Evidence for Alfvén waves powering auroral arc via a static electric potential drop. *Nature comms.* 17, 297 (2026).
4. Xu, Y., **Yao, Z. H. ***, Arridge, C. S., Zhang, B., Chen, J. J., Badman, S. V., ... & Wei, Y. (2026). Dawn-dusk Asymmetrical Distribution of Saturn's Cusp. *Nature comms.*, 17(1), 1861.
5. Gu, W. D., Wei, Y., Yao, Z. H., Wang, Y. Q., and Yuan, C. J. (2026). Jupiter wind as a unifying bridge: a testable paradigm in the Tianwen-4 era. *Earth Planet. Phys.*, 10(4), 1–11. DOI: 10.26464/epp2026051
6. Gu, W., Wei, Y.*, **Yao, Z.***, Yan, L., Xu, Y., & Zhang, B. (2026). An investigation on the apparent north-south asymmetry identified in multiple current sheet crossing events in Jovian dawn outer magnetosphere. *Journal of Geophysical Research: Space Physics*, 131(3), e2025JA034597.
7. Sun, J. W., Xie, L., **Yao, Z. H.***, Fu, S. Y., Zhang, B., Xu, Y., ... & Dunn, W. R. (2025). Determining the State of Jupiter's Magnetosphere Using In Situ and Remote Sensing Observations. *The Astrophysical Journal*, 995(1), 7.
8. Wang T Y, Zhang H S*, **Yao Z H***, Zhang B Z, Feng E H, Fu J Y. 2025. Theoretical analysis of interchange instability in Jupiter's inner magnetosphere[J]. *Reviews of Geophysics and Planetary Physics*, 56(0): 1-13 (in Chinese). doi:10.19975/j.dqyxx.2025-013.
9. Shang, Z., **Yao, Z.***, Liu, J., Xu, L., Xu, Y., Zhang, B., ... & Wei, Y. (2025). Statistical study of auroral variability under different solar wind conditions based on classification using deep learning techniques. *Earth and Planetary Physics*, 9(6), 1163-1170.
10. Duanmu, X., Ye, S., & **Yao, Z.*** (2025). Wave Responses to Magnetosheath Boundary Motion at Saturn: Mirror Modes and Electromagnetic Ion Cyclotron Waves. *The Astrophysical Journal*, 991(2), 213.
11. Teng, S., Zhang, J., **Yao, Z.***, & Xiao, P. (2025). Analysis of Ion Cyclotron Waves during Cassini's Flybys of Enceladus. *The Astrophysical Journal*, 989(1), 45.
12. **Yao, Z. H.**, Xu, Y., Zeng, Z. L., Zhang, B., Grodent, D., Bonfond, B., et al. (2025). Rotating finger-like structures of Jovian magnetodisc. *Geophysical Research Letters*, 52, e2025GL116277. <https://doi.org/10.1029/2025GL116277>
13. Zeng Z, **Yao Z***, Liu J, et al. Ultralow-frequency Waves in Jupiter's Magnetopause Boundary Layer[J]. *The Astrophysical Journal*, 2024, 976(1): 92.
14. Xu, Y., Arridge, C. S., **Yao, Z. H.***, Zhang, B., Ray, L. C., Badman, S. V., ... & Wei, Y. (2024). In situ evidence of the magnetospheric cusp of Jupiter from Juno spacecraft measurements. *Nature comms.*, 15(1), 6062.
15. Gu W D, Wei Y, **Yao Z H**, Gao J W, Wang Y Q, He F, Rong Z J. 2024. Research on Jupiter's

magnetic field: Review and prospect. *Reviews of Geophysics and Planetary Physics*, 55(0): 1-14 (in Chinese). doi:10.19975/j.dqyxx.2023-060.

16. Xu, Y., Ray, L., **Yao, Z.***, Zhang, B., Bonfond, B., Badman, S., ... & Wei, Y. (2024). Revealing the local time structure of the Alfvén radius in Jupiter's magnetosphere through high-resolution simulations. *Journal of Geophysical Research: Planets*, 129(6), e2024JE008368.
17. Pan, D. X., **Yao, Z. H.***, Guo, R. L., Arridge, C. S., Ray, L. C., Zhao, Y., ... & Bolton, S. J. (2024). The Colocation of Magnetic Reconnection and Current Disruption in Jovian Middle Magnetosphere. *The Astrophysical Journal Letters*, 969(2), L35.
18. Sun J W, Xie L, **Yao Z H***, et al. On the global features of the 10–60-min ULF waves in Jovian magnetosphere: Juno observations[J]. *Journal of Geophysical Research: Planets*, 2024, 129(4): e2023JE008279.
19. **Yao Z H**, Guo R L, Wei Y, et al. Recent advances in the magnetic reconnection, dipolarization, and auroral processes at giant planets from the perspective of comparative planetology[J]. *Earth and Planetary Physics*, 2024, 8(5): 659-672.
20. Gu, W. D., **Yao, Z. H.***, Wei, Y., Qin, T. S., Zhang, B. Z., Xu, Y., ... & Chen, Y. N. (2024). A survey of magnetic field line curvature in Jovian dawn magnetodisc. *Geophysical Research Letters*, 51(1), e2023GL106971.
21. Zhang, B., **Yao, Z.***, Brambles, O. J., Delamere, P. A., Lotko, W., Grodent, D., ... & Lyon, J. G. (2024). A unified framework for global auroral morphologies of different planets. *Nature Astro.*, 1-9
22. Xu, Y., **Yao, Z. H.***, Ye, S. Y., Badman, S. V., Dialynas, K., Sergis, N., ... & Wei, Y. (2023). A Possible Unified Picture for the Connected Recurrent Magnetic Dipolarization, Quasi-Periodic ENA Enhancement, SKR Low-Frequency Extension and Narrowband Emission at Saturn. *Journal of Geophysical Research: Space Physics*, 128(9), e2023JA031445.
23. Xu, Y., **Yao, Z. H.***, Zhang, B., Delamere, P. A., Ray, L. C., Dunn, W. R., ... & Wei, Y. (2023). On the relation between Jupiter's aurora and the dawnside current sheet. *Geophysical Research Letters*, 50(13), e2023GL104123.
24. Duanmu, X. Y., **Yao, Z. H.***, Wei, Y., and Ye, S. Y. (2023). Two types of mirror mode waves in the Kronian magnetosheath. *Earth Planet. Phys.*, 7(3), 414–420 doi: 10.26464/epp2023040
25. Gu, W. D., **Yao, Z. H.***, Pan, D. X., Xu, Y., Zhang, B., Delamere, P. A., et al. (2023). Hourly periodic variations of ultralow-frequency (ULF) waves in Jupiter's magnetosheath. *Journal of Geophysical Research: Planets*, 128, e2022JE007625.
26. Shang, Z., **Yao, Z. H.***, Liu, J., Xu, L., Xu, Y., Zhang, B., Guo, R. and Wei, Y. (2023) Automated Classification of Auroral Images with Deep Neural Networks. *Universe*, 9, 96.
27. **Yao, Z. H.**, Bonfond, B., Grodent, D., Chané, E., Dunn, W. R., Kurth, W. S., et al. (2022). On the relation between auroral morphologies and compression conditions of Jupiter's magnetopause: Observations from Juno and the Hubble Space Telescope. *Journal of Geophysical Research: Space Physics*, 127, e2021JA029894.
28. Zhao, Y., Yan, L., **Yao, Z.***, et al. Auroral responses to the visit of comet 73P/Schwassmann–Wachmann 3 in 2006. *Geosci. Lett.* 9, 36 (2022). <https://doi.org/10.1186/s40562-022-00245-8>
29. **Yao, Z H**, Sun, T R. 2022. Origin of Jupiter's soft X-ray aurora and a brief perspective on future explorations. *Reviews of Geophysics and Planetary Physics*, 53(0): 1-2. doi:10.19975/j.dqyxx.2021-066
30. Guo, R. L., **Yao, Z. H.***, Grodent, D., Bonfond, B., Clark, G., Dunn, W. R., et al. (2021). Jupiter's double-arc aurora as a signature of magnetic reconnection: Simultaneous observations from HST and Juno. *Geophysical Research Letters*, 48, e2021GL093964.
31. Guo, R. L., **Yao, Z. H.***, Dunn, W. R., Palmaerts, B., Sergis, N., Grodent, D., ... & Dougherty, M. K. (2021). A rotating azimuthally distributed auroral current system on Saturn Revealed by the Cassini spacecraft. *The Astrophysical Journal Letters*, 919(2), L25.
32. Xu, Y., Guo, R. L., **Yao, Z. H. *** et al., (2021) Properties of plasmoids observed in Saturn's dayside and nightside magnetodisc. *Geophysical Research Letters*, 48, e2021GL096765
33. **Yao, Z. H.**, Dunn, W. R. et al., (2021) Revealing the source of Jupiter's x-ray auroral flares.

Science Advances, 7, eabf0851.

34. **Yao Z H**, Guo R L, Yuan C J, Pan D X, Xu Y. (2021) Recent advances in giant planetary space environments. *Reviews of Geophysics and Planetary Physics*, 52(5): 543-560
35. Pan, D-X., **Yao, Z-H.***, Manners, H., Dunn, W., Bonfond, B., Grodent, D., et al. (2021). Ultralow-frequency waves in driving Jovian aurorae revealed by observations from HST and Juno. *Geophysical Research Letters*, 48, e2020GL091579
36. Bonfond*, B., **Yao, Z. H.*** (equal contribution), Gladstone, G. R., Grodent, D., Gérard, J.C., Matar, J., et al. (2021). Are dawn storms Jupiter's auroral substorms?. *AGU Advances*, 2, e2020AV000275. <https://doi.org/10.1029/2020AV000275>
37. Zhang, B.*, Delamere, P. A., **Yao, Z. H.***(correspondence) et al. (2021) How Jupiter's Unusual Magnetospheric Topology Structures Its Aurora, *Science Advances*, doi: 10.1126/sciadv.abd1204
38. **Yao, Z. H.**, Bonfond, B., Clark, G., Grodent, D., Dunn, W. R., & Vogt, M. F., et al. (2020). Reconnection- and dipolarization-driven auroral dawn storms and injections. *Journal of Geophysical Research: Space Physics*, 125, e2019JA027663. <https://doi.org/10.1029/2019JA027663>
39. Pan, D-X., **Yao, Z-H. ***(correspondence), Guo, R-L., Bonfond, B., Wei, Y., Dunn, W., et al. (2020). A Statistical Survey of Low-Frequency Magnetic Fluctuations at Saturn. *Journal of Geophysical Research: Space Physics*, 125, e2020JA028387. <https://doi.org/10.1029/2020JA028387>
40. **Yao, Z. H.**, D. Grodent, W. S. Kurth, G. Clark, B. H. Mauk, T. Kimura, B. Bonfond, S.-Y. Ye, A. T. Lui, A. Radioti, B. Palmaerts, W. R. Dunn, L. C. Ray, F. Bagenal, S. V. Badman, I. J. Rae, R. L. Guo, Z. Y. Pu, J.-C. Gérard, K. Yoshioka, J. D. Nichols, S. J. Bolton and S. M. Levin (2019), On the relation between Jovian aurorae and the loading/unloading of the magnetic flux: simultaneous measurements from Juno, HST and Hisaki, *Geophysical Research Letters*, doi: 10.1029/2019GL084201
41. Radioti, A., **Zhonghua Yao***(correspondence), Denis Grodent, Benjamin Palmaerts, E Roussos, Kostas Dialynas, D Mitchell, Z Pu, SV Badman, J-C Gérard, W Pryor, Bertrand Bonfond (2019), Auroral beads at Saturn and the driving mechanism: Cassini proximal orbits, *The Astrophysical Journal Letters*, doi: 10.3847/2041-8213/ab4e20
42. **Yao, Z. H.**, Aikaterini Radioti, D Grodent, Licia C Ray, B Palmaerts, N Sergis, Kostas Dialynas, AJ Coates, Christopher Stephen Arridge, E Roussos, Sarah Victoria Badman, Sheng-Yi Ye, J-C Gérard, PA Delamere, RL Guo, ZY Pu, JH Waite, N Krupp, Donald G Mitchell, MK Dougherty (2018), Recurrent magnetic dipolarization at Saturn: revealed by Cassini, *J. Geophys. Res. Space Physics*, doi: 10.1029/2018JA025837
43. Guo, R. L., **Z. H. Yao***(correspondence) et al. (2018), Reconnection acceleration in Saturn's dayside magnetodisc: a multicase study with Cassini, *The Astrophysical Journal Letters*, doi: 10.3847/2041-8213/aaedab/pdf
44. Guo, R. L., **Z. H. Yao***(correspondence) et al. (2018), Rotationally driven magnetic reconnection in Saturn's dayside, *Nature Astronomy*, doi: 10.1038/s41550-018-0461-9
45. Wei, Y., **Z. H. Yao***(correspondence), W. X. Wan (2018), China's roadmap for planetary exploration, *Nature Astronomy*, doi: 10.1038/s41550-018-0456-6
46. **Yao, Z. H.**, K. R. Murphy, I. J. Rae and N. Balan (2017), Introduction to the thematic series coupling of the Magnetosphere-Ionosphere system. *Geoscience Letters*, doi: 10.1186/s40562-017-0092-5.
47. **Yao, Z. H.**, A. Radioti, I. J. Rae, J. Liu, D. Grodent, L. C. Ray, S. V. Badman, A. J. Coates, J. -C. Gerard, J. H. Waite, J. N. Yates, Q. Q. Shi, Y. Wei, B. Bonfond, M. K. Dougherty, E. Roussos, N. Sergis and B. Palmaerts (2017), Mechanisms of Saturn's near-noon transient aurora: in situ evidence from Cassini measurements. *Geophysical Research Letters*, doi:10.1002/2017GL075108.
48. **Yao, Z. H.**, A. J. Coates, L. C. Ray, I. J. Rae, D. Grodent, G. H. Jones, M. K. Dougherty, C. J. Owen, R. L. Guo, W. R. Dunn, A. Radioti, Z. Y. Pu, G. R. Lewis, J. H. Waite and J.-C. Gerard (2017), Co-rotating magnetic reconnection site in Saturn's magnetosphere. *The Astrophysical Journal Letters*, doi: 10.3847/2041-8213/aa88af.

49. **Yao, Z. H.**, I. J. Rae, A. T. Lui, K. R. Murphy, C. J. Owen, Z. Y. Pu, C. Forsyth, D. Grodent, Q. G. Zong, A. M. Du and N. Kalmoni (2017), An explanation of auroral intensification during the substorm expansion phase. *J. Geophys. Res. Space Physics*, doi: 10.1002/2017JA024029.
50. **Yao, Z. H.**, D. Grodent, L. C. Ray, I. J. Rae, A. J. Coates, Z. Y. Pu, A. T. Lui, A. Radioti, J. H. Waite, G. H. Jones, R. L. Guo and W. R. Dunn (2017), Two fundamentally different drivers of dipolarizations at Saturn, *J. Geophys. Res. Space Physics*, 122, 4348–4356
51. **Yao, Z. H.**, I. J. Rae, R. L. Guo, A. N. Fazakerley, C. J. Owen, R. Nakamura, W. Baumjohann, C. E. J. Watt, K. J. Hwang, B. L. Giles, C. T. Russell, R. B. Torbert, A. Varsani, H. S. Fu, Q. Q. Shi and X.-J. Zhang (2017), A direct examination of the dynamics of dipolarization fronts using MMS, *J. Geophys. Res. Space Physics*, 122, 4335–4347
52. **Yao, Z. H.**, Z. Y. Pu, I. J. Rae, A. Radioti, and M. V. Kubyshkina (2017), Auroral streamer and its role in driving wave-like pre-onset aurora, *Geoscience Letters*, 4(1), 8.
53. **Yao, Zhonghua** (2017), Observations of loading-unloading process at Saturn distant magnetotail. *Earth Planet. Phys.*, 1, 53-57. <http://doi.org/10.26464/epp2017007>
54. **Yao, Z. H.**, A. N. Fazakerley, A. Varsani, I. J. Rae, C. J. Owen, D. Pokhotelov, C. Forsyth and R. L. Guo (2016), Substructures within a dipolarization front revealed by high-temporal resolution Cluster observations, *J. Geophys. Res. Space Physics*, 121, doi:10.1002/2015JA022238
55. **Yao, Z. H.**, J. Liu, C. J. Owen, C. Forsyth, I. J. Rae, Z. Y. Pu, H. S. Fu, X. -Z. Zhou, Q. Q. Shi, A. M. Du, R. L. Guo and X. N. Chu (2015), A physical explanation for the magnetic decrease ahead of dipolarization fronts. *Annales Geophysicae*, doi:10.5194/angeo-33-1301-201.
56. **Yao, Z. H.**, Z. Y. Pu, C. J. Owen, S. Y. Fu, X. N. Chu, J. Liu, V. Angelopoulos, I. J. Rae, C. Yue, X. -Z. Zhou, Q. -G. Zong, X. Cao, Q. Q. Shi, C. Forsyth and A. M. Du (2014), Current Reduction in a Pseudo-breakup Event: THEMIS Observations. *J. Geophys. Res. Space Physics*, 119, doi: 10.1002/2014JA020186
57. **Yao, Z.H.**, Z. Y. Pu, A. M. Du, V. Angelopoulos, C. J. Owen, J. Liu, X. N. Chu, X. Cao, S. Y. Fu, Q. -G. Zong and Y. Wang (2014), Pressure Gradient Evolution in the Near-Earth Magnetotail at the Arrival of BBFs. *Chin. Sci. Bull.*, 59, doi: 10.1007/s11434-014-0618-6.
58. **Yao, Z. H.**, V. Angelopoulos, Z. Y. Pu, S. Y. Fu, M. Kubyshkina, J. Liu, X. N. Chu, T. Nishimura, X. Cao, A. M. Du, C. Yue, Q. Q. Shi and Y. Wei (2013), Conjugate observations of flow diversion in the magnetotail and auroral arc extension in the ionosphere. *J. Geophys. Res. Space Physics*, 118, doi:10.1002/jgra.50419.
59. **Yao, Z. H.**, W. J. Sun, S. Y. Fu, Z. Y. Pu, J. Liu, V. Angelopoulos, X. -J. Zhang, X. N. Chu, Q. Q. Shi, R. L. Guo and Q. -G. Zong (2013), Current Structures Associated with Dipolarization Fronts. *J. Geophys. Res. Space Physics*, 118, doi:10.1002/2013JA019290
60. **Yao, Z. H.**, Z. Y. Pu, S. Y. Fu, V. Angelopoulos, M. Kubyshkina, X. Xing, L. Lyons, Y. Nishimura, L. Xie, X. G. Wang, C. J. Xiao, X. Cao, J. Liu, H. Zhang, M. Nowada, Q. -G. Zong, R. L. Guo, J. Zhong and J. X. Li (2012), Mechanism of substorm current wedge formation: THEMIS observations. *Geophysical Research Letters*. 39(13): p. L13102.